SMA-ACC network in strategic action sequencing

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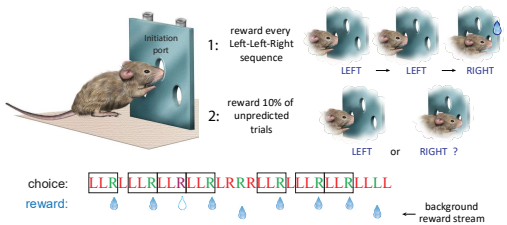
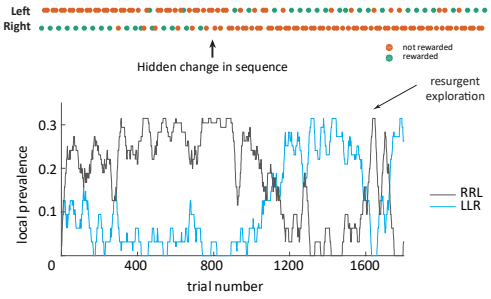
Abstract

Functional imaging studies in humans have revealed that under the conditions when subjects are forced to base their choices on limited or ambiguous data, the strongest changes in neural activity occur in the medial frontal cortex (MFC) – a collection of cortical regions that includes anterior cingulate cortex (ACC) as well as the associated medial premotor cortices: pre- supplementary motor area (pre-SMA), supplementary motor area (SMA) and the cingulate motor areas (CMA). Previous work from our lab has contributed to the emerging view that the ACC plays a key and selective role in generating, evaluating, and refining the nascent internal model—a set of hypotheses about the causal rules of the environment and the best behavioral strategies—under conditions of uncertainty. Intriguingly, a recent study (Bonini et al, 2014) posited that SMA may instead play a leading role in strategy selection. Based on a timing analysis of the EGG signals recorded in humans, the authors suggested that SMA may be hierarchically upstream of ACC. However, whether the observed EEG signal reflects a local computation in the SMA or the timing of its major input, and whether SMA activity affects behavior by shaping computations in the ACC remains unclear. Here we functionally map a likely rat homologue of the SMA, and provide evidence from anatomical tracings, physiological recordings and targeted perturbation experiments in support of the proposed hierarchy within the SMA-ACC circuit.

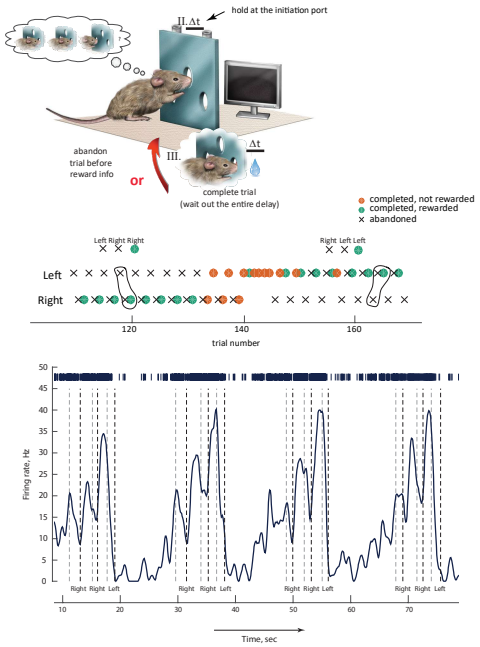
Task that exposes strategic action sequencing



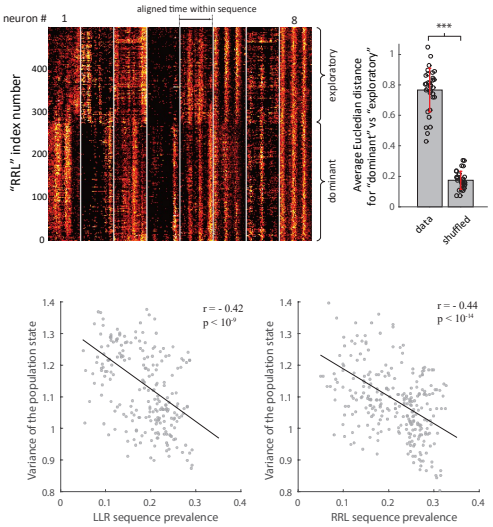
An animal has to discover that a particular target sequence of 'Left' (L) and 'Right' (R) actions (nose pokes at response ports) is preferentially rewarded. After initiating a trial at the central port, an animal is eligible to receive a reward only if his choice of the reward port either differs from that predicted by the computer or is the last in a series of choices that conforms to a specific (latent) target sequence of actions (here, 'LLR'). Thus, the animal has to discover not only that a particular target sequence of actions is preferentially rewarded, but do so in the presence of a parallel reward stream that weakly encourages unpredictable rather than structured behavior by providing alternative hypotheses that need to be evaluated before the consistently rewarded sequence can be discovered.



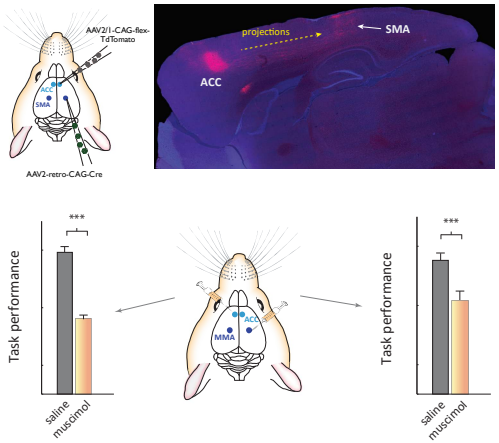
Rats indeed strategically sequence their actions



Internally-driven changes in strategy prevalence as an organizing principle of ACC ensemble dynamics

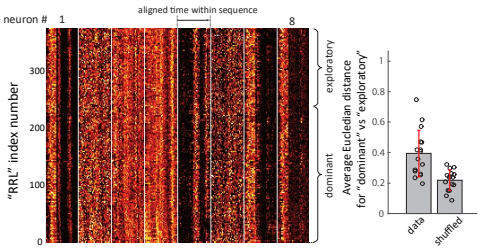


Anatomical and functional mapping of a putative rat homologues of the SMA

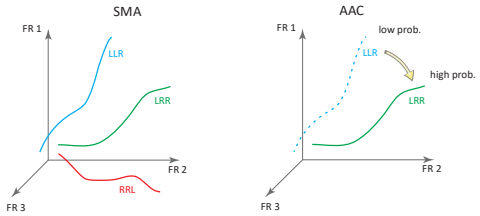


Following a bilateral injection of muscimol into either ACC or putative SMA, the animals continue to make choices (and thereby performing basic 'initiation port - side port' sequencing) for hundreds of trials, but are no longer able to implement strategic higher-order sequencing to conform their choices to the preferentially rewarded target pattern.

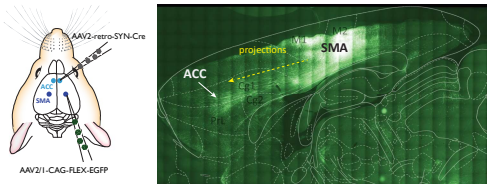
Stability of sequence-related representation in SMA across behavioral contexts



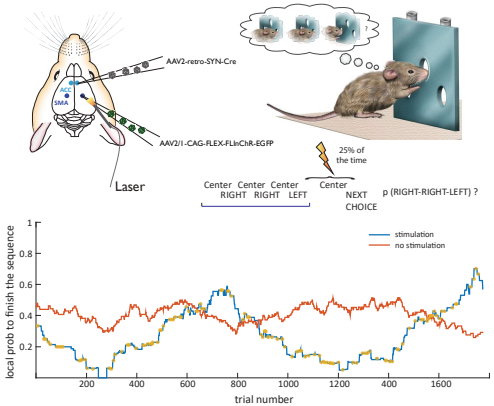
A basis set for a probabilistic model in ACC?



SMA provides a strong input to ACC



Silencing SMA-ACC projection at the time of strategy selection affect task performance



Conclusions

We uncovered strong sequence-related responses and exposed one revealing organizational principle of ACC network dynamics: a marked modulation of neural representation for individual sequences based on their prevalence. The most striking modulation is based on whether a sequence represents the dominant strategy, or is evaluated in a transient burst of resurgent exploration, with an additional modulation by the local prevalence of that sequence within any behavioral epoch. Importantly, such marked shifts in encoding do not appear to be a brain-wide phenomenon. Indeed, even in a presumed homologue of the supplementary motor area (SMA) activity associated with a sequence remains stable across contexts.

The stability of sequence representation in SMA suggests an intriguing possibility that SMA provides a stable basis for sequence representation that ACC utilizes in different ways depending on the inferred behavioral context and the internally-guided need for further exploration.

Aknowledgements

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